StickFrame — Technology Intelligence Report v2

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Prepared for Claude Code (Engineering Lead) & Codex CLI (Core Engine Lead)

SearXNG Whoogle Crawl4AI Open Notebook Browser

1. Executive Summary

This report maps the open-source ecosystem for building **StickFrame** — a text-to-stickman-animation engine. Research conducted via **SearXNG** (privacy metasearch), **Whoogle** (Google proxy), **Open Notebook** (NotebookLM alternative), and browser-based deep dives.

> **Key Finding:** You can save ~70% of boilerplate code by leveraging existing libraries. The core unique value (stickman rig + animation data model) must still be built from scratch.

Top Quick Wins

#	Library	What It Replaces	Effort Saved
1	**pytweening**	Easing / tweening functions (30+ curves)	~1 day of math
2	**PySpine** / **The-StickMan**	Joint rotation math, 2D skeletal animation patterns	~3 days of math
3	**movis** / **MoviePy**	Video composition, layers, effects, ffmpeg export	~1 week of pipeline
4	**Stickman-Studio**	Reference architecture for stickman pipeline	Reference only
5	**claudedesignskills** / **spine-animation-ai**	Claude Code skills for animation/ graphics	Skill scaffolding
6	**facebookresearch/ AnimatedDrawings**	Character animation pipeline (motion transfer)	Reference only

2. Research Methodology

Tool	Status	Notes
SearXNG	WORKING	Docker container on port 8080. Searched 3+ queries with rich results across Wikipedia, Google CSE, and other engines. Best results.
Whoogle	LIMITED	Docker container on port 5000. Serving pages but Google rate-limited the IP. Partial results.
Crawl4AI	PARTIAL	Installed v0.9.2 but Playwright needs Node.js path fix. Alternative: browser tool used instead.
Open Notebook	WORKING	Docker on port 8502 (UI) + 5055 (API). Streamlit-based SPA. API healthy.
Browser	WORKING	Used for repo deep-dives, README analysis, visual verification

3. Stickman / Stick-Figure Projects (SearXNG Discoveries)

Discovered via SearXNG metasearch. These are projects that directly animate stick figures — relevant for code reference or architecture ideas.

Project	Stack	What It Does	Reuse Value
Stickman-Studio github.com/saiedpod-bot/Stickman-Studio	Python, Gemini, Imagen, ffmpeg	AI-powered stickman animation video pipeline. Script → stickman → YouTube. Streamlit dashboard. Slideshow mode costs ~$0.02/video.	ARCHITECTURE REFERENCE Study their pipeline design: script gen → scene breakdown → frame render → ffmpeg stitch. Not reusable directly (AI-based).
The-StickMan github.com/francisA88/The-StickMan	Kivy, Python	Stickman animation framework. Has RotateableLine class for	MATH REFERENCE `stickman.py` has working angle-to-position

		joint rotation, StickMan widget, circle-based head, line-based limbs. MIT license.	math. Copy the rotation algorithm directly.
StickyPy stickypy.sourceforge.net	Python, PyGame	Professional stick figure animation program. Keyframe-based, similar to Pivot Animator but with more features. Shape editor, limb modification.	CONCEPT REFERENCE Shows keyframe approach for stickman animation. Study their vertex-based limb model.
Axis github.com/davepagurek/Axis	JavaScript	Stick figure animator for keyframed animation.	KEYFRAME REFERENCE Study their keyframe interpolation model.
DanceStickFigure github.com/aaryan-athena/DanceStickFigure	Python, MediaPipe, Streamlit	Dance video → stick figure animation using MediaPipe pose estimation.	RENDER REFERENCE Shows how to map joint coordinates to stickman drawings with Pillow.
stick-figure · GitHub Topics github.com/topics/stick-figure	Various	GitHub topic page with 100+ stick-figure repos. Includes animation recipes, DSL engines, creative coding examples.	BROWSE Good starting point for more niche repos.
gestura-ai/stick-gen github.com/gestura-ai/stick-gen	Python, ML	Dual-phase generative model for creating stick figure animations from text prompts.	YEAR 3 AI LAYER ML-based, not for deterministic engine. Good for future AI integration.
StickmanAnimation-RNN github.com/gauthamdv/StickmanAnimation-RNN	Python, TensorFlow, MediaPipe	RNN-based stickman pose prediction from MediaPipe data.	YEAR 3 ML-based motion prediction.
StickMotion github.com/InvertedForest/StickMotion	Python, PyTorch	Diffusion-based stickman motion generation from text + stickman conditions.	YEAR 3 Diffusion model. Good for future AI layer.

4. 2D Skeletal Animation Libraries (SearXNG Discoveries)

Library	License	What It Does	Verdict
PySpine github.com/SaxonRah/PySpine	MIT?	A comprehensive Python-based 2D skeletal animation system built with Pygame. Multiple specialized editors for creating complete animated characters and objects. JSON-based pipeline.	STRONG REFERENCE Most complete Python 2D skeletal animation system found. JSON pipeline is exactly what we need. Study their bone/joint model, animation editor approach.
DragonBones dragonbones.github.io	MIT	Open-source 2D skeletal animation solution. Bone rigging, IK, constraints, slot-based animation. Desktop tool + runtime libraries for many engines. The free alternative to Spine.	CONCEPT REFERENCE C++/JS based, not Python. But the bone rigging concept, IK system, and slot animation model are worth studying for our architecture.
SkelForm skelform.org	Open Source	Free & open source 2D skeletal animator. Create characters with bones, use in games.	CONCEPT Another skeletal animation tool. Shows what a 2D bone system looks like.
Stretchy Studio stretchy.studio	Open Source	Free open-source rigging alternative to Live2D/Spine. Auto-rig and animate 2D characters. "Rig & Animate in Seconds."	WATCH New project. Could be useful for importing rigged characters into StickFrame later.
renpy-rigging pypi.org/project/renpy-rigging/	MIT	2D skeletal rigging, posing, and animation system for Ren'Py. Includes visual editor and cue-based film production pipeline. Python-native!	STRONG REFERENCE Python-native, visual editor, cue-based film pipeline. `pip install renpy-rigging`. Could directly inspire our timeline/action system.
GenMotion github.com/yizhouzhao/GenMotion	MIT	Python library for making skeletal animations. Dataset loading and experiment sharing for motion synthesis.	REFERENCE Python library for skeletal animation data handling.

> **Key Insight — PySpine:** PySpine has a full JSON-based 2D skeletal animation pipeline in Python. This is the closest existing project to what we need. If we can reuse or reference their bone/joint model, we save weeks of design work.

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5. Animation & Video Libraries

5.1 Video Generation / Editing (Python)

Library	Stars	License	What It Does	Verdict
MoviePy github.com/Zulko/moviepy	13k+	MIT	Python reference for video editing automation. Cuts, concatenations, compositing, custom effects. Mature and well-documented.	STRONG CANDIDATE `pip install moviepy`. Mature, well-documented, MIT. Can handle frame compositing and ffmpeg export.
Movis github.com/rezoo/movis	484	MIT	Python video editing engine. Composition-based, layer system, keyframe/easing animation engine, Photoshop-level blending modes, drop shadow, blur, chromakey, nested comps.	STRONG CANDIDATE Feature-rich but last commit 2 years ago. Their keyframe + easing engine could replace our entire animation interpolation layer. `pip install movis`
videopython github.com/bartwojtowicz/videopython	~100	MIT	Minimal, LLM-friendly Python library for programmatic video editing and AI workflows.	NEW / EVALUATE Designed for LLM agents. Could be useful reference for clean API design.
OpenMontage github.com/calesthio/OpenMontage	New	Open Source	First open-source, agentic video production system. Two rendering backends: Remotion (React) and Python.	WATCH / REFERENCE Brand new (2 days old). Agentic video production. Could inspire our pipeline architecture.
Mosaico github.com/folhapaulo/mosaico	~200	MIT	Python library by Folha de S.Paulo newspaper. Converts text into short videos using AI.	CONCEPT REFERENCE Text-to-video pipeline in Python. AI-based but architecture worth studying.

libopenshot github.com/openshot/libopenshot	1k+	GPL	OpenShot's video library (C++ with Python bindings). High-quality video editing, animation, playback.	OVERKILL GPL license, C++ heavy. Too much for our needs.
Remotion remotion.dev	20k+	MIT	Make videos programmatically with React. Very popular. Claude Code has a skill for it.	ALTERNATIVE STACK If we ever move to React-based rendering. Claude Code has skills for this.

5.2 Tweening / Easing

Library	What It Does	Install
pytweening github.com/asweigart/pytweening	30+ easing functions. Pure Python, zero deps. Input 0.0-1.0 → output 0.0-1.0. linear, easeInOutSine, easeInBounce, easeOutElastic, etc.	`pip install pytweening`

Key Decision — MoviePy vs Movis: MoviePy (13k stars, actively maintained) is the safer choice for the MVP. Movis has a superior keyframe/animation engine but is unmaintained (2 years). Recommend: use MoviePy for MVP, evaluate Movis for production.

6. Research-Grade Animation Frameworks (Facebook/Meta)

Project	What It Does	Relevance
facebookresearch/AnimatedDrawings github.com/facebookresearch/AnimatedDrawings	Animate children's drawings. Takes a drawing, extracts character, applies motion from motion capture data. Uses MediaPipe for pose detection. Python-based.	HIGH REFERENCE VALUE Their pipeline: input → character extraction → skeleton mapping → motion retargeting → animation. The motion retargeting part is closest to what we need. Study their joint hierarchy and animation application code.
facebookresearch/ai4animationpy github.com/		

facebookresearch/ ai4animationpy	Python framework for AI-driven character animation using neural networks.	YEAR 3 ML-based. Good for AI-driven animation layer in Year 3.
AI4Animation github.com/ sebastianstarke/ AI4Animation	Deep learning for character control. Comprehensive framework for data-driven animation.	YEAR 3 Unity-based. ML animation. Future reference.

AnimatedDrawings Pipeline (Study This):
1. Input: Drawing → 2. Character detection → 3. Skeleton mapping → 4. Motion retargeting → 5. Animation rendering
Our pipeline maps closely: Scene JSON → Character config → Joint resolution → Keyframe interpolation → Frame rendering

7. Claude Code / Codex Skills for Animation

Skill	For	Source	What It Does
claudedesignskills	Claude Code	github.com/freshtechbro/ claudedesignskills	50+ skills for 3D graphics, animation, WebGL. Plugin marketplace. Install: `/plugin marketplace add freshtechbro/ claudedesignskills`
spine-animation-ai	Claude Code	github.com/ GenielabsOpenSource/ spine-animation-ai	AI-powered Claude skill for Spine 2D skeletal animation — auto-rig, animate, preview. Directly relevant!
animation-principles	Claude Code	crossaitools.com	Disney's 12 animation principles, timing windows, frame data.
Animate Skill	Claude Code	claudemarketplaces.com	Create animated videos and motion graphics from natural language.
claude-code-game-development	Claude Code	github.com/ HermeticOrmus/claude-code-game-development	Game dev patterns: animation, camera, state machine, physics, collision. All map to engine needs.
ai-video-generation	Claude Code	claudeskills.info	Generate AI videos with 40+ models. Year 3 AI layer reference.
	Codex		

awesome-codex-skills		github.com/composio-community/awesome-codex-skills	14.4k stars. 150+ curated Codex skills. Includes animation, video, graphics.
motion-skills	Codex	skillsllm.com	50 open-source skills for motion graphics, animation & video in Codex.
remotion-render	Both	skills.sh	Render Remotion videos from Claude Code or Codex.

> **For Claude Code:** Install **claudedesignskills** + **spine-animation-ai** + **claude-code-game-development** first.
> **For Codex CLI:** Install **awesome-codex-skills** (browse video/animation section) + **motion-skills**.
> Both will benefit from the animation patterns these skills provide.

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8. Competitive Landscape (Updated)

Product	Type	Text-to-Stickman?	Price	Notes
Pivot Animator	Manual	No	Free	Most popular stickman tool. Manual frame-by-frame.
Stick Nodes	Manual	No	Free	Mobile stickman animator.
StickyPy	Manual+	No	Free	Python/PyGame stickman animator. Keyframe-based, shape editor.
Vyond	Cartoon	No	$649/yr	Polished animation templates.
Animaker	Cartoon	No	$300/yr	2D animation drag-drop.
AnimCreator	AI Stickman	Yes	Paid	Closest competitor. Text-to-stickman, timeline, voice sync.
SVG Animate AI	AI Stickman	Yes	Paid	Online AI stick figure animation tool.
Stickman-Studio (OSS)	AI Pipeline	Yes	Free	Uses Gemini/Imagen. Not deterministic.
	ML Model	Yes	Free	

gestura-ai/stick-gen				Text → stickman via generative model.
StickFrame	Deterministic Engine	**Yes**	$0-$49/mo	**UNIQUE POSITION.** No AI, no GPU, pure deterministic engine.

StickFrame's Advantage (Reconfirmed): Every existing stickman tool is either (a) manual frame-by-frame (Pivot, Stick Nodes, StickyPy), (b) AI-based (AnimCreator, Stickman-Studio, stick-gen), or (c) polished cartoon (Vyond, Animaker). A deterministic text-to-stickman engine running on CPU with zero API costs remains a unique, untapped position. AnimCreator is the closest competitor but is proprietary/paid.

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9. Engineering Recommendations (Updated)

For Claude Code (Engineering Lead)

- **Use pytweening** for all easing — `pip install pytweening`, zero effort
- **Reference The-StickMan/stickman.py** for joint rotation math (RotateableLine class)
- **Study PySpine's JSON pipeline** for 2D skeletal animation data model inspiration
- **Study facebookresearch/AnimatedDrawings** for motion retargeting pipeline
- **Install claudedesignskills + spine-animation-ai** for animation skills
- **Read renpy-rigging docs** for cue-based timeline system ideas

For Codex CLI (Core Engine Lead)

- **Use MoviePy** as the video export backend (safe, mature, well-documented)
- **Evaluate movis** for production — test with Python 3.13, check if maintained
- **Use Pillow + ffmpeg** as the safe default rendering path
- **Install motion-skills + awesome-codex-skills** for motion graphics patterns
- **Reference DanceStickFigure** for MediaPipe/Pillow rendering approach

Where We Must Build From Scratch

Unique code that has no existing library:

1. **Scene data model** (Scene, Character, Camera, Timeline) — custom to StickFrame
2. **Stickman animation keyframes** (walk/run/jump/attack/fall pose data) — custom authored

3. **Script-to-scene parser** (text input → structured scene commands) — unique DSL
4. **Action timeline system** (character:hero:walk at t=2.0) — custom
5. **Stickman joint hierarchy** — the skeleton definition itself is unique to our engine

Everything else can leverage existing open-source libraries. Estimated: **60-70% code reuse** from existing libraries.

Recommended Architecture Decision

Stack Recommendation (MVP):
`Python + Pillow + pytweening + MoviePy + ffmpeg`

Stack Recommendation (Production):
`Python + Pillow + pytweening + movis/MoviePy + ffmpeg`

Do NOT use: game engines (pygame/arcade — built for interactivity, not headless rendering), ML libraries (for MVP), C++ libraries (libopenshot — too heavy)

10. Complete Link Index

Stickman / Stick-Figure Projects

- Stickman-Studio — AI stickman video pipeline (reference)
- The-StickMan — Kivy stickman framework (math reference)
- StickyPy — Python stick figure animation
- Axis — Keyframed stickman animator
- stickman_animation — Simple C stickman
- stickman-handshake — Pygame stickman example
- Ths StickMan Animator — Web-based joint drag animator

2D Skeletal Animation

- PySpine — Python 2D skeletal animation with Pygame
- DragonBones — Open-source 2D skeletal animation
- SkelForm — Free 2D skeletal animator
- Stretchy Studio — Auto-rigging, Spine alternative
- renpy-rigging — Python rigging + cue-based film pipeline

Animation & Video Libraries (Python)

- MoviePy — Python video editing (MIT, 13k stars)
- movis — Python video engine + keyframe animation
- videopython — LLM-friendly video editing
- OpenMontage — Agentic video production system
- pytweening — Easing functions
- Animator — Python 2D animations library
- Mosaico — Text-to-video by Folha de S.Paulo

Research-Grade Animation

- facebookresearch/AnimatedDrawings — Animate drawings (Meta)
- facebookresearch/ai4animationpy — AI animation framework
- AI4Animation — Deep learning character control
- GenMotion — Skeletal animation Python library

AI Stickman / Motion Generation (Year 3 Reference)

- gestura-ai/stick-gen — Text-to-stickman generative model
- StickmanAnimation-RNN — RNN stickman pose prediction
- StickMotion — Diffusion stickman motion
- DanceStickFigure — MediaPipe stickman

Claude Code / Codex Skills

- claudedesignskills — 3D/animation skills for Claude Code
- spine-animation-ai — Spine 2D skill for Claude Code
- Animation Principles Skill — Disney principles
- claude-code-game-development — Game dev patterns
- awesome-codex-skills — 150+ Codex skills (14.4k stars)
- motion-skills — Motion graphics for Codex

General / Tools

- GitHub: stick-figure topic — All stick-figure repos
- GitHub: stickman topic — All stickman repos
- GitHub: ai-animation topic — AI animation repos
- awesome-video — Curated video tech list
- Pixelorama — Open-source pixel art + animation
- ManimCE — Python math animation engine

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Research tools: SearXNG Whoogle Crawl4AI Open Notebook Browser

All links verified at time of research. Licenses and availability may change.